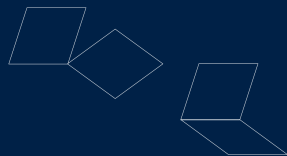


Freiman's theorem - Proof and Best Bounds

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Plan for today

- ▶ Recall a few things - what is Freiman's Theorem?
- ▶ Necessary tools for the proof of Freiman's Theorem:
 - ▶ Plünnecke-Ruzsa Inequality & Ruzsa Covering Lemma
 - ▶ Freiman Homomorphisms and Isomorphisms
 - ▶ Ruzsa Modeling Lemma
 - ▶ Bohr sets and Bogolyubov's Lemma
 - ▶ Geometry of Numbers
- ▶ Proof of the theorem
- ▶ Bounds conjectured and best bounds proved so far



Definition 1

A **generalized arithmetic progression** (GAP) of **dimension** d is a set of the form:

$$\{x_0 + l_1x_1 + \dots + l_dx_d \mid 0 \leq l_1 < L_1, \dots, 0 \leq l_d < L_d \in \mathbb{Z}\}$$

where $x_0, x_1, \dots, x_d \in \mathbb{Z}$ and $L_1, L_2, \dots, L_d \in \mathbb{N}$.

- ▶ **Size** of a GAP: $L_1L_2\dots L_d$.
- ▶ If there are no coincidences of elements, then the GAP is called **proper**.



Doubling Constant - Recall

Definition 2

The doubling constant of a finite subset, A , of an abelian group is the ratio $\frac{|A+A|}{|A|}$

We have seen that:

- ▶ APs have doubling constant at most 2.
- ▶ Proper GAPs have doubling constant at most 2^d .
- ▶ Any subset of an AP has a bounded doubling constant:
If B is a finite arithmetic progression and $A \subseteq B$ has $|A| \geq C|B|$, then
 $|A+A| \leq |B+B| \leq 2|B| \leq \frac{2}{C}|A| \implies A$ has doubling constant at most $\frac{2}{C}$.
- ▶ The same property holds for GAPs as well.
- ▶ Now we ask the converse question - if a set has bounded doubling constant, must it be contained in a GAP? - Freiman's Theorem answers this question!!



Freiman's Theorem - Recall

Theorem 3

If $A \subseteq \mathbb{Z}$ is a finite set and $|A + A| \leq K|A|$, then A is contained in a GAP of dimension at most $d(K)$ and size at most $f(K)|A|$, where $d(K)$ and $f(K)$ are constants depending only on K .

- ▶ This theorem basically says that any finite set of integers lies in a GAP which has constant dimension and whose size is a factor of the size of A !
- ▶ The conclusion of the theorem can be obtained for a proper GAP as well. It comes at the cost of increasing $d(K)$ and $f(K)$. (Tao and Vu, 2006)

Necessary tools for the proof of Freiman's Theorem



- ▶ Plünnecke-Ruzsa Inequality & Ruzsa Covering Lemma
- ▶ Freiman Homomorphisms and Isomorphisms
- ▶ Ruzsa Modeling Lemma
- ▶ Bohr sets and Bogolyubov's Lemma
- ▶ Geometry of Numbers



Plünnecke-Ruzsa Inequality & Ruzsa Covering Lemma

Plünnecke-Ruzsa Inequality

Theorem 4

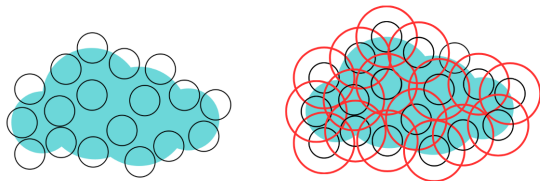
If A is a finite subset of an abelian group and $|A + A| \leq K|A|$, then

$$|mA - nA| \leq K^{m+n}|A|$$

Ruzsa Covering Lemma

Theorem 5

Let X and B be subsets of an abelian group. If $|X + B| \leq K|B|$, then there exists a subset $T \subset X$ with $|T| \leq K$ such that $X \subset T + B - B$.





Freiman Homomorphisms and Isomorphisms

Definition 6

Let A, B be subsets in (possibly different) abelian groups. We say that $\phi : A \rightarrow B$ is a **Freiman s -homomorphism** (or a Freiman homomorphism of order s), if

$$\phi(a_1) + \dots + \phi(a_s) = \phi(a'_1) + \dots + \phi(a'_s)$$

whenever $a_1, \dots, a_s, a'_1, \dots, a'_s \in A$ satisfy

$$a_1 + \dots + a_s = a'_1 + \dots + a'_s$$

Definition 7

If $\phi : A \rightarrow B$ is a bijection, and both ϕ and ϕ^{-1} are Freiman s -homomorphisms, then ϕ is said to be a **Freiman s -isomorphism**.



Freiman Homomorphisms and Isomorphisms

Examples:

1. Any group homomorphism is a Freiman s -homomorphism for any order, s .
2. If A is a group without any additive structure, (for example, $\{1, 10, 100, 1000\}$), and $B = \mathbb{Z}$, then any map, $\phi : A \rightarrow B$ is a Freiman 2-homomorphism.
3. $\{0, 1\} \rightarrow \mathbb{Z}/2\mathbb{Z}$ is a Freiman 2-homomorphism, and is bijective, but is not a Freiman 2-isomorphism.
4. In general, $\mathbb{Z} \rightarrow \mathbb{Z}/N\mathbb{Z}$ is a Freiman homomorphism of every order, but not a Freiman isomorphism.

Proposition 1

If $A \subset \mathbb{Z}$ has diameter smaller than N/s , then $(\text{mod } N)$ maps A Freiman s -isomorphically to its image.



Remarks:

- ▶ If two sets are Freiman 2-isomorphic, then APs are preserved between the two sets!
- ▶ Observe that if a subset B is Freiman 8-isomorphic to A' , then $2B - 2B$ is Freiman 2-isomorphic to $2A' - 2A'$.
- ▶ Hence in our proof, it is sufficient that we look at Freiman 8-isomorphisms in the set $2B - 2B$.



Ruzsa Modeling Lemma

Theorem 8

Let $A \subset \mathbb{Z}$, $s \geq 2$, and N be a positive integer such that $N \geq |sA - sA|$. Then there exists $A' \subset A$ with $|A'| \geq |A|/s$ such that A' is Freiman s -isomorphic to a subset of $\mathbb{Z}/N\mathbb{Z}$.

Idea of Proof.

- Let $q > \max(sA - sA)$ be a prime. For every $\lambda \in [q - 1]$, consider

$$\phi : \mathbb{Z} \rightarrow \mathbb{Z}/q\mathbb{Z} \xrightarrow{\lambda} \mathbb{Z}/q\mathbb{Z} \rightarrow [q].$$

For all λ , there exists an interval, $I_\lambda \subset [q]$ of length less than q/s such that $A_\lambda = \{a \in A : \phi(a) \in I_\lambda\}$ has more than $|A|/s$ elements. ϕ , when restricted to this A_λ is a Freiman s -homomorphism.

- Now consider $\psi : \mathbb{Z} \xrightarrow{\phi} [q] \rightarrow \mathbb{Z}/N\mathbb{Z}$

By some additional arguments, we can show that there is some λ such that ψ maps A_λ s -isomorphically to its image. We simply take $A' = A_\lambda$.



Ruzsa Modeling Lemma

Let $A \subset \mathbb{Z}$, $s \geq 2$, and N be a positive integer such that $N \geq |sA - sA|$. Then there exists $A' \subset A$ with $|A'| \geq |A|/s$ such that A' is Freiman s -isomorphic to a subset of $\mathbb{Z}/N\mathbb{Z}$.

Corollary 9

If $A \subset \mathbb{Z}$ with $|A + A| \leq K|A|$, then there exists a prime $N \leq 2K^{16}|A|$ and some $A' \subset A$ with $|A'| \geq |A|/8$ such that A' is Freiman 8-isomorphic to a subset of $\mathbb{Z}/N\mathbb{Z}$.

Proof.

$|8A - 8A| \leq K^{16}|A|$ (by Plünnecke Ruzsa Inequality).

Choose a prime, N , such that $K^{16}|A| \leq N < 2K^{16}|A|$.

Apply modeling lemma with $s = 8$ to get the required result. □



Bohr sets and Bogolyubov's Lemma

Definition 10

Suppose $R \subset \mathbb{Z}/N\mathbb{Z}$. Define $\text{Bohr}(R, \epsilon) = \{x \in \mathbb{Z}/N\mathbb{Z} : \|\frac{rx}{N}\|_{\mathbb{R}/\mathbb{Z}} \leq \epsilon, \text{ for all } r \in R\}$
 $|R|$ is the dimension of the set, denoted by d , and ϵ is called the width of the set.

Bogolyubov's Lemma

Theorem 11

If $A \subset \mathbb{Z}/N\mathbb{Z}$ and $|A| = \alpha N$, then $2A - 2A$ contains some Bohr set, $\text{Bohr}(R, 1/4)$, with $|R| < 1/\alpha^2$.

Proof.

- ▶ $1_{2A-2A}(x) = \sum_{r \in \mathbb{Z}/N\mathbb{Z}} |\hat{1}_A(r)|^4 \cos(\frac{2\pi rx}{N})$
- ▶ For $R := \{r \in \mathbb{Z}/N\mathbb{Z} \setminus \{0\} \mid |\hat{1}_A(r)| > \alpha^{1.5}\}$, if $x \in B(R, 1/4)$, then $1_{2A-2A}(x) > 0$, and hence $x \in 2A - 2A$.





Geometry of Numbers

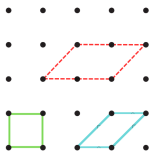
Definition 12

A **lattice** in $\mathbb{R}^d$ is a set given by $\Lambda = \mathbb{Z}v_1 \oplus \dots \oplus \mathbb{Z}v_d$, where $v_1, \dots, v_d \in \mathbb{R}^d$ are linearly independent vectors.

Definition 13

The **determinant** $\det(\Lambda)$ of a lattice $\Lambda = \mathbb{Z}v_1 \oplus \dots \oplus \mathbb{Z}v_d$ is the absolute value of the determinant of the matrix with $v_1, \dots, v_d$ as columns.

Remark: The determinant of a lattice is also equal to the volume of the fundamental parallelepiped.





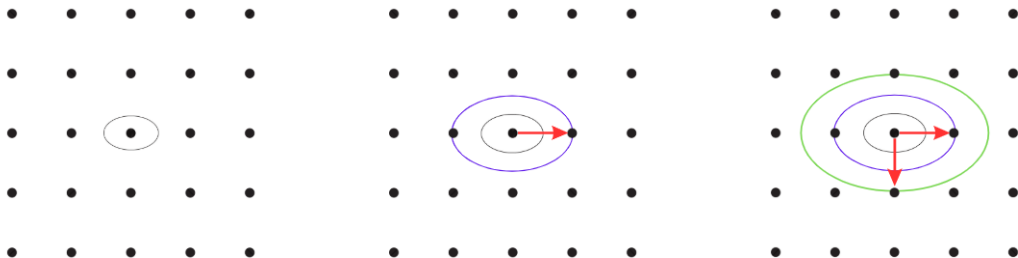
Definition 14

Given a centrally symmetric convex body $K \subset \mathbb{R}^d$ (centrally symmetric: $x \in K \iff -x \in K$), define the i^{th} successive minimum of K with respect to a lattice Λ as

$$\lambda_i = \inf\{\lambda \geq 0 : \dim(\text{span}(\lambda K \cap \Lambda)) \geq i\}$$

for $1 \leq i \leq d$. Equivalently, λ_i is the minimum λ such that λK contains i linearly independent lattice vectors from Λ .





A directional basis of K with respect to Λ is a basis $b_1, \dots, b_d$ of $\mathbb{R}^d$ such that $b_i \in \lambda_i K$ for each $i = 1, \dots, d$.



Minkowski's second theorem

Theorem 15

Let $\Lambda \in \mathbb{R}^d$ be a lattice and K a centrally symmetric convex body. Let $\lambda_1 \leq \dots \leq \lambda_d$ be the successive minima of K with respect to Λ . Then,

$$\lambda_1 \dots \lambda_d \operatorname{vol}(K) \leq 2^d \det(\Lambda)$$

- We have seen that $2A - 2A$ contains a Bohr set. We will now show that every Bohr set contains a proper GAP!



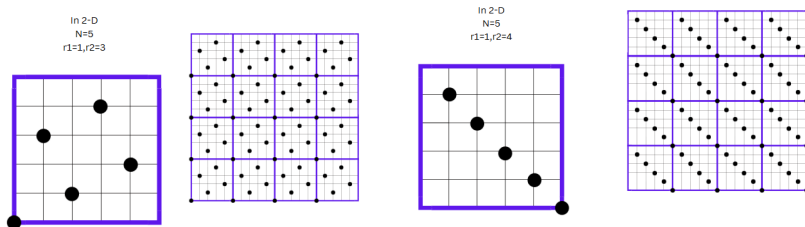
Geometry of Numbers

Theorem 16

Let N be a prime. Every Bohr set of dimension d and width $\epsilon \in (0, 1)$ in $\mathbb{Z}/N\mathbb{Z}$ contains a proper GAP with dimension at most d and size at least $(\epsilon/d)^d N$.

Proof.

Let $R = \{r_1, \dots, r_d\}$, $v = (\frac{r_1}{N}, \dots, \frac{r_d}{N})$. Consider the lattice $\Lambda \subset \mathbb{R}^d$ consisting of all points in $\mathbb{R}^d$ that are congruent mod 1 to some integer multiple of v .





Geometry of Numbers

Proof.

Note that $\det(\Lambda) = 1/N$.

Consider $K = [-\epsilon, \epsilon]^d$. Let $\lambda_1, \dots, \lambda_d$ be the successive minima of K with respect to Λ .

Let $b_1, \dots, b_d$ be the directional basis.

By construction, $\|b_j\|_\infty \leq \lambda_j \epsilon$ for all $1 \leq j \leq d$.

For each j , let $L_j = \lceil 1/(\lambda_j d) \rceil$. $0 \leq l_j < L_j \implies \|l_j b_j\|_\infty < \frac{\epsilon}{d}$.

Hence, $\|l_1 b_1 + \dots + l_d b_d\|_\infty \leq \epsilon$.

Since each b_j is $x_j v$ plus a vector with integer coordinates for some $0 \leq x_j < N$, we

have $\left\| \frac{(l_1 x_1 + \dots + l_d x_d) r_i}{N} \right\|_{\mathbb{R}/\mathbb{Z}} \leq \epsilon$ for all i .

Hence the GAP, $\{l_1 x_1 + \dots + l_d x_d : 0 \leq l_i \leq L_i \text{ for all } i\}$ is in $\text{Bohr}(R, \epsilon)$.

We can show that it's size is large using Minkowski's theorem. We also need to show that it is proper, which we can do by similarly bounding the norm of

$b = (l_1 - l'_1) b_1 + \dots + (l_d - l'_d) b_d$ by ϵ , forcing b to be 0.





► Corollary of Ruzsa Modeling Lemma

If $A \subset \mathbb{Z}$ with $|A + A| \leq K|A|$, then there exists a prime $N \leq 2K^{16}|A|$ and some $A' \subset A$ with $|A'| \geq |A|/8$ such that A' is Freiman 8-isomorphic to a subset of $\mathbb{Z}/N\mathbb{Z}$.

► Results related to Freiman's Isomorphisms

1. If two sets are Freiman 2-isomorphic, then APs are preserved between the two sets.
2. If a subset B is Freiman 8-isomorphic to A' , then $2B - 2B$ is Freiman 2-isomorphic to $2A' - 2A'$.

► Bogolyubov's Lemma

If $A \subset \mathbb{Z}/N\mathbb{Z}$ and $|A| = \alpha N$, then $2A - 2A$ contains some Bohr set, $\text{Bohr}(R, 1/4)$, with $R < 1/\alpha^2$.



► Result from Geometry of Numbers

Let N be a prime. Every Bohr set of dimension d and width $\epsilon \in (0, 1)$ in $\mathbb{Z}/N\mathbb{Z}$ contains a proper GAP with dimension at most d and size at least $(\epsilon/d)^d N$

► Ruzsa Covering Lemma

Let X and B be subsets of an abelian group. If $|X + B| \leq K|B|$, then there exists a subset $T \subset X$ with $|T| \leq K$ such that $X \subset T + B - B$.

► Plünnecke-Ruzsa inequality

If A is a finite subset of an abelian group and $|A + A| \leq K|A|$, then $|mA - nA| \leq K^{m+n}|A|$



Proof of Freiman's Theorem

Proof.

- ▶ Start with a set A with $|A + A| \leq K|A|$. By the corollary of the Ruzsa Modeling Lemma, we map $A' \subset A$ (with $|A'| \geq |A|/8$) to $B \subset \mathbb{Z}/N\mathbb{Z}$, where N is a prime such that $N \leq 2K^{16}|A|$, and A' and B are 8-Freiman isomorphic.
- ▶ Take $\alpha = \frac{|B|}{N} = \frac{|A'|}{N} \geq \frac{|A|}{8N} \geq \frac{1}{16K^{16}}$.
Bogolyubov's Lemma $\implies 2B - 2B$ contains a Bohr set, $\text{Bohr}(R, 1/4)$, where $|R| < 1/\alpha^2 \leq 256K^{32}$
- ▶ By the geometry of numbers, there is a proper GAP inside this Bohr set, with $d < 256K^{32}$ and size $\geq (\frac{\epsilon}{d})^d N = (4d)^{-d} N$.
- ▶ This GAP is mapped to a proper GAP, say Q , in $2A' - 2A'$ (by properties of Freiman's isomorphisms), with same dimension and size.



Proof of Freiman's Theorem

Proof.

► $Q \subset 2A - 2A \implies Q + A \subset 3A - 2A.$

By Plünnecke-Ruzsa inequality, $|Q + A| \leq |3A - 2A| \leq K^5|A|$

► $|Q| \geq (4d)^{-d}N$ and $N \geq |A'| \geq |A|/8$
 $\implies |Q + A| \leq K'|Q|$

► By Ruzsa Covering Lemma, $\exists X \subset A$, with $|X| \leq K'$, such that $A \subset X + Q - Q$.

► X is contained in a GAP with dimension $|X|$ with length 2 in each direction, and $Q - Q$ is a GAP with same dimension, d , as Q .

Hence, $X + Q - Q$ is contained in a GAP of

dimension $\leq |X| + d \leq K' + d \leq d(K),$

and size $\leq 2^{|X|+d}|Q| \leq 2^{K'+d}|2A - 2A| \leq 2^{K'+d}K^4|A| \leq f(K)|A|$





Bounds in the theorem - $f(K)$ and $d(K)$

- ▶ With the above proof, we get the bounds: $d(K) = e^{K^{O(1)}}$ and $f(K) = e^{e^{K^{O(1)}}}$.
- ▶ Chang(2002) showed that by modifying the covering lemma part of the proof, we can get polynomial bounds $d(K) = K^{O(1)}$ and $f(K) = e^{K^{O(1)}}$.
- ▶ Sanders(2012) proved the bounds, $d(K) = K(\log K)^{O(1)}$ and $f(K) = e^{K(\log(K))^{O(1)}}$, which are the best bounds obtained so far.
- ▶ Conjectured bounds are $d(K) = O(K)$ and $f(K) = 2^{O(K)}$.



- ▶ Lecture Videos: Graph Theory and Additive Combinatorics | Prof. Yufei Zhao | Fall 2023 | MIT OpenCourseWare
- ▶ Graph Theory and Additive Combinatorics | Yufei Zhao